ChronoLens

Unveiling Temporal Mysteries

What if a photograph could transport you into a forgotten version of history?

A narrative adventure that turns the historic centre of Rome into a snow-bound mystery — built from 18th-century engravings, modelled in Rhino and played in Unreal Engine.

DISCIPLINE
Interactive & 3D

YEAR
January 2024

CONTEXT
Sapienza University of Rome — Computer Graphics and Gaming Studio

ROLE
Game concept, narrative, research, 3D modelling, Unreal levels, coding, GDD

TOOLS
Rhinoceros 3D · CAD · Unreal Engine 4.27 / 5.2 · Adobe suite

WITH
Farnaz Attar



02 — BEFORE YOU BEGIN

A game told in five acts

ChronoLens is a game design project: a story, a researched city, a set of hand-modelled buildings and a playable level. This file follows it in that order — the idea first, then the research behind every street, then how Rome was rebuilt in 3D, and finally how it plays.

The Game Design Document it is based on runs to more than sixty pages. Here it is condensed into sixteen boards.

Engravings & photos

Research references, credited as in the original document.

White renders

Clay models straight out of Rhinoceros, before texturing.

Colour frames

Captured in Unreal Engine — the game as it plays.

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03 — AT A GLANCE

A history lesson disguised as a crime

Nerezza, a history student, takes a photograph with an old camera and wakes up in a distorted, snow-covered Rome. To get home she must solve a photographer's murder — and every clue sends her through real streets, real buildings and real history, with the Pantheon at the centre of it all.



GENRE	Narrative action-adventure, casual
PLATFORM	PC · Microsoft Windows
STRUCTURE	Linear levels; tasks to complete before Nerezza can return to her own time
MISSION	Teach players about 18th-century Rome and the Pantheon through play
PLAYER FEELING	The curiosity and empathy of a student — or simply a history lover
PLAYERS	From middle school to university, and fans of story-led, atmospheric games
PARTNERS	Museums and universities that want to make Rome's history more engaging
PRICE	€15–20 · students €10–15
ACCESSIBILITY	Subtitles for all dialogue, captions for sound effects

04 — THE STORY

One photograph, two Romes

Written as a detective story so that the history is never a lecture: every place the player must visit is a place worth knowing.



I Summer

On a hot June day Nerezza wanders an old bazaar, follows a trail of strange picture frames into an antique shop and flees an angry old man — with his camera in her hands. She photographs a frame. Drops of blood appear. Everything fades to black.



II Winter

She wakes in a cold, distorted Rome; even the Pantheon's facade has changed. A young girl who cannot remember her name becomes her guide, and explains the city's custom: when someone is killed, a mannequin takes their place. A photographer has been murdered. Nobody knows by whom.



III The frames

Across the city Nerezza finds hidden picture frames, each holding a fragment of one photograph. Piece by piece the image reveals what happened — and who did it.



IV The fire

She confronts the murderer — a priest serving in the Pantheon. He tries to destroy the evidence and attack her; in the chaos a fire takes the building, and him with it.

EPILOGUE *She wakes in the summer heat again. The camera is gone, the frames are full of bougainvillea — and the photographer's spirit can finally rest.*



05 — THE CAST

Nerezza

Italian for "darkness" — a fitting name for a time traveller.

A 23-year-old history student and the player's stand-in. Clever and ironic, with a very Roman humour that plays down even the most dramatic moments — which lets her, and the player, look at each mystery from a clear distance.

Curious | Adventurous | Visionary | Restless | Freedom-loving | Clever | Ironic

DIALOGUE DESIGN

In conversation she answers simply; her real voice comes through triggered interface moments, where her own observations reveal who she is.



The young girl

Cannot remember her own name. Nerezza helps her find it; in return she becomes Nerezza's guide.



The city's people

Strange inhabitants who hold clues — and a murderer hiding in plain sight.

06 — THE SITE

A square kilometre of Rome

The game is set in the Campus Martius, around the Pantheon. Before modelling anything, the area was mapped twice: on an atlas of ancient Rome and on today's streets — to fix which buildings the player would meet, and where.

01 Pantheon — the heart of the game

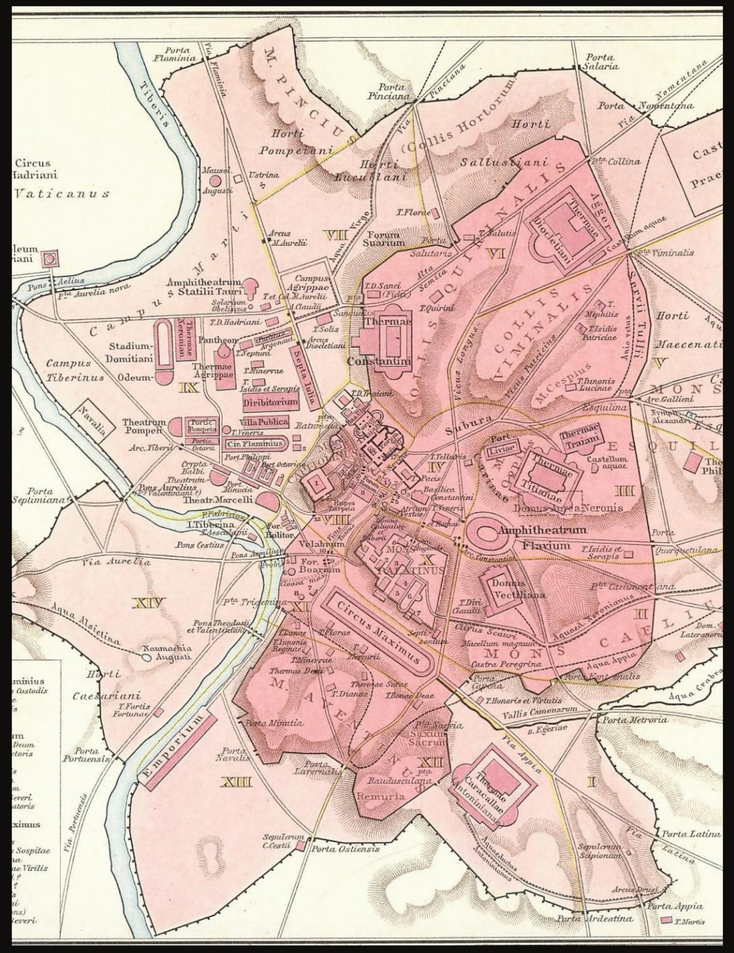
02 Palazzo Montecitorio

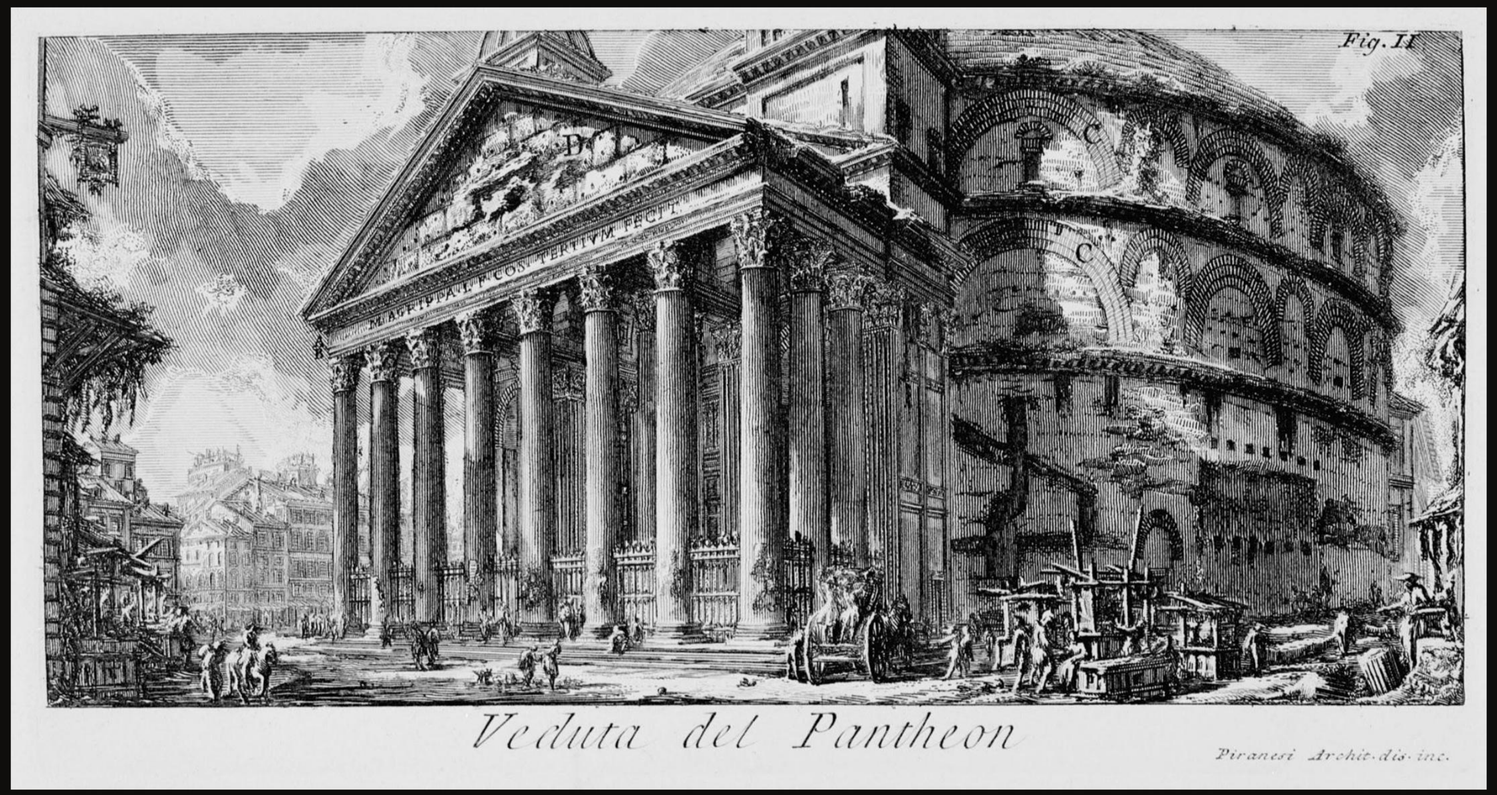
03 Chiesa di Sant'Ignazio di Loyola

04 Piazza di Pietra

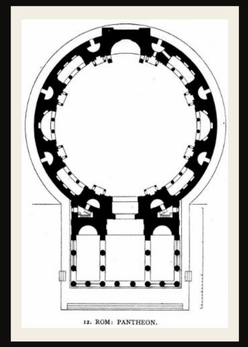
05 Piazza Colonna

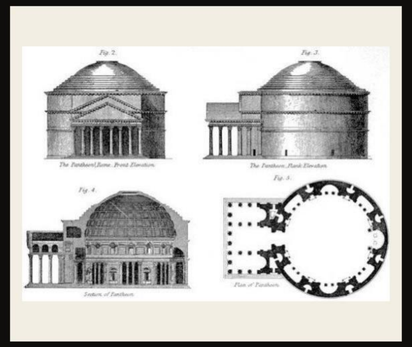
06 The fountain in Piazza della Rotonda


Porta Flaminia
Porta Pinciana
Porta Salaria
Horti Pompeiani
Horti Lucullani
Collis Hortorum
Horti Sallustiani
Circus Hadriani
Vaticanus
Campus Martius
Amphitheatrum Statilii Tauri
Stadium Domitiani
Campus Tiberinus
Odeum
Thermae Agrippae
Pantheon
Thermae Constantini
Collis Quirinalis
Collis Viminalis
Collis Esquilinus
Collis Palatinus
Theatrum Pompei
Villa Publica
Diribitorium
Theatr. Marcelli
Via Aurelia
Via Flaminia
Via Lata
Via Nomentana
Via Salaria
Via Appia
Via Ostiensis
Via Latina
Via Labicana
Via Praenestina
Via Tiburtina
Via Triumphalis
Via Cornelia
Via Septimiana
Via Aurelia Nova
Via Alsietina
Via Portuensis
Via Ardeatina
Via Laurentina
Via Ostiensis
Via Appia Nova
Via Latina
Via Labicana
Via Praenestina
Via Tiburtina
Via Nomentana
Via Salaria
Via Flaminia
Via Cornelia
Via Triumphalis



Giovanni Battista Piranesi, Veduta del Pantheon









07 — THE PANTHEON

The anchor of the story

A temple to all the gods, a church, a tomb — and in ChronoLens the starting point, the hiding place of the murderer and the scene of the climax. It was studied through engravings, plans and sections before a single line was drawn.

29–19 BC	Built by Marcus Agrippa as part of his complex on the Campus Martius
2nd c.	Rebuilt under Hadrian, who kept Agrippa's inscription on the pediment
609 AD	Consecrated as a church — Santa Maria ad Martyres
1623–44	Pope Urban VIII adds two bell towers, as in the 18th-century views

08 — THEN & NOW

Every building, read twice

Each landmark on the route was compared across time — the 18th- and 19th-century view against the street today — to decide what the game's Rome should look like: old enough to feel strange, familiar enough to recognise.



Sant'Ignazio di Loyola

Baroque church built 1626–1650, dedicated to the founder of the Jesuits.

Palazzo Montecitorio

Seat of the Chamber of Deputies; designed by Bernini for Cardinal Ludovico Ludovisi.



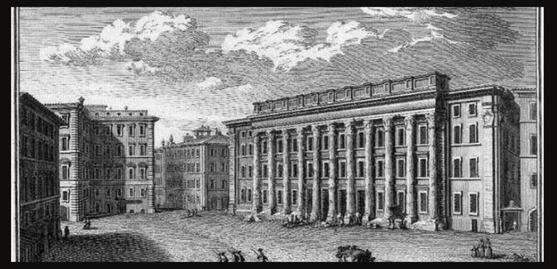
1752



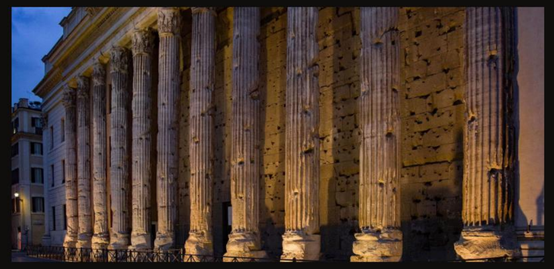
Today

Piazza di Pietra

The Temple of Hadrian, dedicated by Antoninus to his deified father.



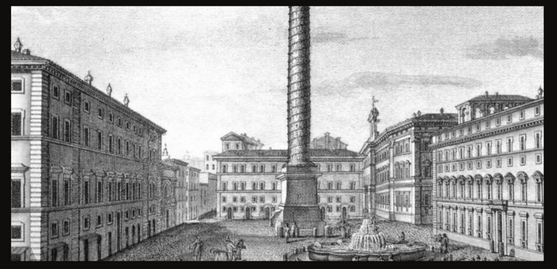
1752



Today

Piazza Colonna

Named for the Column of Marcus Aurelius, standing since AD 193.



19th century

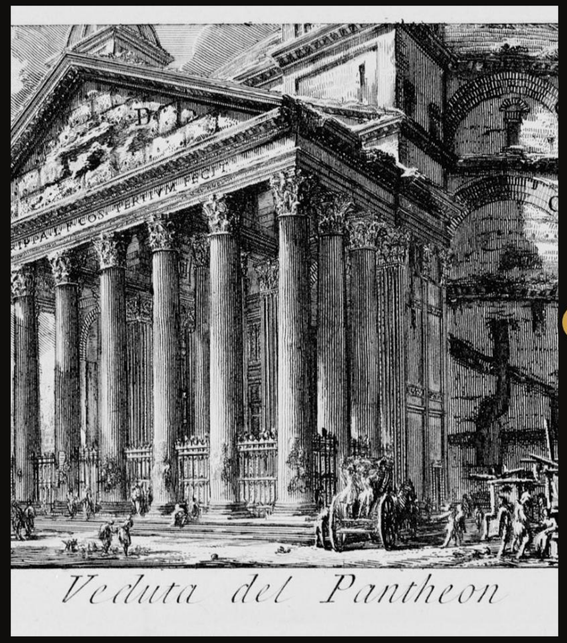


Today

09 — ENGRAVING TO ENGINE

From paper to playable

The Pantheon shows the method used for every building in the game: read the history, model it exactly, then let the story change it.



01 Reference

18th-century engravings fix the proportions and the bell towers.



02 Model

Every element built by hand in Rhinoceros 3D, checked against plans and sections.



03 Material

Weathered stone and brick applied, ready for the engine.



04 Scene

Placed in Unreal Engine, under snow, at the centre of the level.





Urban VIII's bell towers



Portico columns and railings



The rotunda from behind

10 — DETAILS

Down to the letters

Corinthian capitals, coffered pediment, iron railings and the inscription itself — modelled, not painted on — so the building holds up when the player walks right up to it.

ON THE FRIEZE

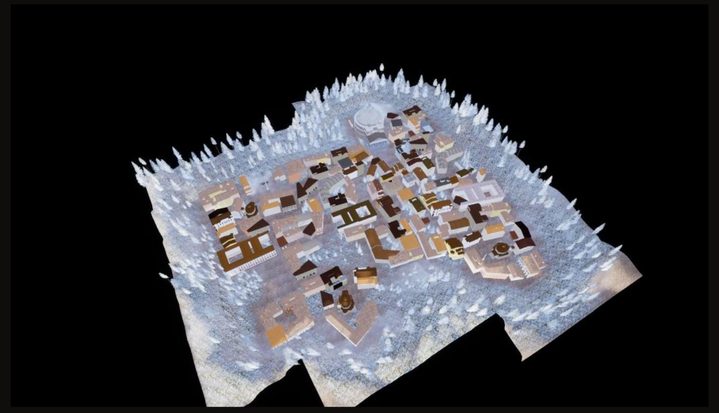
M·AGRIPPA·L·F·COS·TERTIVM·FECIT

“Marcus Agrippa, son of Lucius, consul for the third time, built this.”

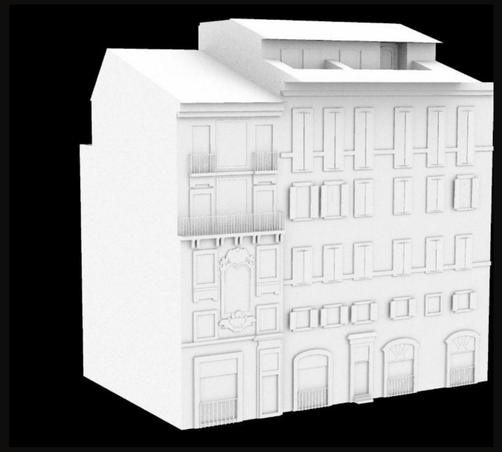
11 — THE CITY

Streets to get lost in

Around the Pantheon, a district of Roman palazzi, a fountain and the everyday objects that make a street believable — and that hide the game's clues.



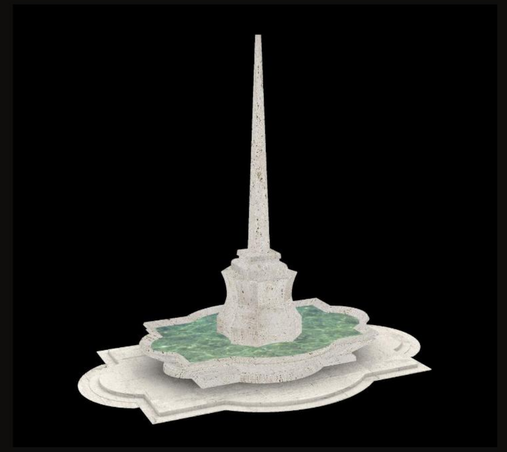
The full level, from above



Palazzo · model



Palazzo · textured



Fountain of the Pantheon

PROPS



Stone bench



Barrel



Wheelbarrow



Inscribed stele



Obelisk



Frames & rubbish



12 — ATMOSPHERE

Rome under snow, and no music

The time jump is told through weather: the story opens in midsummer heat and wakes in a winter that Rome almost never sees. That single contrast tells the player something is wrong before anyone says a word.

Physics use real-world dimensions, and the soundscape stays deliberately bare.

SOUND EFFECTS	Camera shutter · chest opening · collecting stones and pine cones · throwing stones
AMBIENCE	Cold wind blowing · wind hitting the windows
MUSIC	None, on purpose — silence keeps attention on the story and the history
DIALOGUE	Voiced with a voice generator, one voice per character, fully subtitled



BEFORE

25 June · a hot summer day



AFTER

Cold, cloudy, snowing

13 — TUTORIAL LEVEL

Learn every control in one walk



01 Start at the Pantheon. WASD to move, mouse to look.



02 An obstacle — Space to jump.



03 Meet the little girl. T to talk.



04 Follow the path to the crazy man.



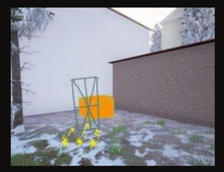
05 He guards a chest: “Get away!”



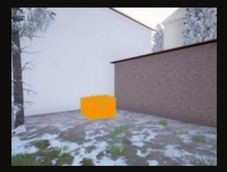
06 Find stones on the ground.



07 C to collect, then go back.



08 E to throw stones at him.



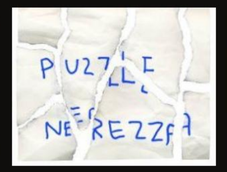
09 He runs off-screen.



10 F to open the chest: torn paper.



11 Open the inventory, select the pieces.



12 The pieces join: the girl's name.



13 Back to the girl. T to talk.



14 Follow the pine cones to the mannequins.

W A S D Move · Space Jump · Mouse Look · photograph · C Collect · T Talk · F Open · E Throw



The route, drawn over the level from above

14 — TRAILER

Thirty shots

A trailer storyboarded shot by shot in the engine: it follows the opening exactly, then switches to questions over gameplay, and ends on the burning Pantheon — showing the hook without spoiling the answer.

- 01–17 · summer: the camera
- 18–26 · winter: the questions
- 27–30 · the fire and the title



01 Logo in the sky; fade; look at the path



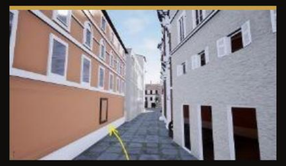
02 Walk to the first frame



03 Frames line the path; walk faster



04 The next frame



05 The next frame



06 On to more frames



07 The final frame; a bench



08 Notice the building



09 Into the wall; fade



10 The camera



11 The old man shouts



12 Out of the antique shop



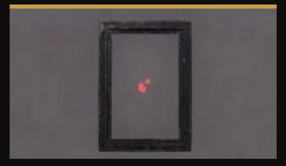
13 Look back; look down at her hands



14 Raise the camera



15 Photograph the frame



16 Drops of blood

17 Black



18 The Pantheon in snow



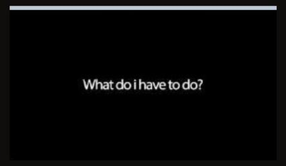
19 “Where am I? What is the date?”



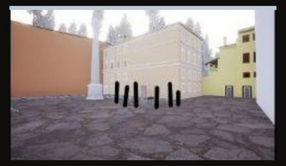
20 The little girl



21 The crazy man



22 “What do I have to do?”



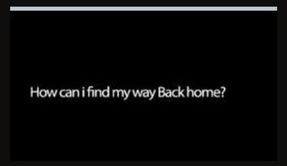
23 The mannequins



24 “Who are these people?”



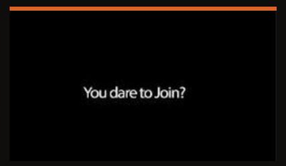
25 Walking the city



26 “How can I find my way back home?”



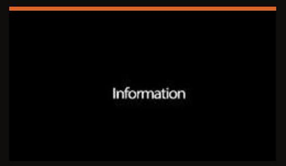
27 The Pantheon burns



28 “You dare to join?”



29 ChronoLens



30 Information



Main menu — Play · Story · Credits · Exit



Story



Settings



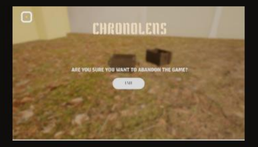
Controls



Audio



General



Exit

15 — GAME UI

Menus that stay in the world

Every screen sits on a blurred, sepia view of the city, so the interface feels like an old photograph laid over the game rather than a panel in front of it.

A single stencil-style wordmark, thin serif headings, rounded keycaps for the controls screen and translucent settings panels for audio — music, effects, ambience, voice — and display options.

16 — CREDITS

The city at peace

The climax burns the very building the game began with. Behind it: a story, a researched city, hand-built models and the code that makes it play — all within one studio course.

ROLE	Game concept · narrative design · historical and visual research · 3D modelling of every building · environment and level development in Unreal Engine · all game coding · game design documentation
TOOLS	Rhinoceros 3D · CAD · Unreal Engine 4.27 and 5.2 · Photoshop · Illustrator · Premiere Pro · After Effects
CONTEXT	Academic · Sapienza University of Rome · Master in Design, Multimedia & Visual Communication · Computer Graphics and Gaming Studio · January 2024
WITH	Farnaz Attar